Patterns & Inductive Reasoning — Practice — Answer Key

A worked example comes before each type. Show your thinking.

Name: _____ Date: _____

Type 1 Find the Next Terms

Watch Me — Worked Example

$5,\ 11,\ 17,\ 23,\ \ldots$ Check every pair: $11-5=6, 17-11=6, 23-17=6$. The rule is **add 6**.
Next two terms: $23+6=29,\ 29+6=35$. **Answer: $29,\ 35$**

1. $4,\ 9,\ 14,\ 19,\ \ldots$ Next two terms?

$24,\ 29$

2. $50,\ 43,\ 36,\ 29,\ \ldots$ Next two terms?

$22,\ 15$

3. $3,\ 6,\ 12,\ 24,\ \ldots$ Next two terms?

$48,\ 96$

4. $1,\ 4,\ 9,\ 16,\ 25,\ \ldots$ Next two terms?

$36,\ 49$

5. $2,\ 3,\ 5,\ 8,\ 12,\ \ldots$ Next two terms?

$17,\ 23$

6. $1,\ -2,\ 4,\ -8,\ \ldots$ Next two terms?

$16,\ -32$

Type 2 Find a Far-Away Term

Watch Me — Worked Example

$7,\ 10,\ 13,\ 16,\ \ldots$ Find the 20th term.
Start at 7 and add 3 **nineteen** times (one fewer than the term number): $7+19\cdot 3=7+57$. **Answer: 64**

7. $2,\ 7,\ 12,\ 17,\ \ldots$ Find the 10th term.

47

8. $100,\ 96,\ 92,\ 88,\ \ldots$ Find the 15th term.

44

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9. Figure 1 has 4 toothpicks, Figure 2 has 7, Figure 3 has 10. How many in Figure 25?

76

10. HARD $1,\ 3,\ 6,\ 10,\ 15,\ \ldots$ Find the 10th term.

55

Type 3 Make a Conjecture

Watch Me — Worked Example

Look at sums of two odd numbers: $1+3=4,\ 5+7=12,\ 9+11=20$.
Every sum is even. **Conjecture: the sum of two odd numbers is even.**

11. $2\cdot 4=8,\ 6\cdot 2=12,\ 10\cdot 8=80$. Make a conjecture about the product of two even numbers.

the product is even

12. $1+3=4,\ 1+3+5=9,\ 1+3+5+7=16$. Conjecture about the sum of the first n odd numbers.

it equals n^2

Type 4 Find a Counterexample

Watch Me — Worked Example

Conjecture: "If a number is divisible by 2, it is divisible by 4."
Try 6: it is divisible by 2, but $6\div 4=1.5$. One counterexample shows the conjecture is false. **Answer: 6**

13. "The square of a number is always greater than the number." Give a counterexample.

$\frac{1}{2}$ **(or** 1**,** 0**)**

14. "Every prime number is odd." Give a counterexample.

2

15. "Two angles that add to 90° are both 45°." Give a counterexample.

30° **and** 60°

16. HARD "n^2+n+1 is prime for every whole number n." Find a counterexample.

$n=4$**:** $21=3\cdot 7$

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